Nunzio Sarnino, DVM, Dipl. ECVPH

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Profile

Veterinarian and European Specialist in Veterinary Public Health (Dipl. ECVPH) with expertise in One Health, antimicrobial resistance (AMR), epidemiology, and Quantitative Microbial Risk Assessment (QMRA). Advanced R user with a track record in large-scale data analysis, decision-support modelling, and risk-assessment model standardisation (FSKX/FAIR). Experienced in coordinating multi-partner data outputs for EU-funded projects and translating complex evidence into decision-relevant deliverables.

Education

Residency in Veterinary Public Health, subspecialty Population Medicine - European College of Veterinary Public Health (ECVPH) | 12/2025

Master's degree in Veterinary Medicine - Università degli Studi di Napoli Federico II, Italy | 03/2018

PhD in Biomedical Sciences - Freie Universität Berlin, Germany | thesis submission expected March 2026

Work Experience

Scientific Research Associate (Short-term) | LMU Munich, Competence Centre for Food Safety, Chair of Food Microbiology, Munich (Germany)
02/2026 – Present

- Analysis of an international survey dataset: data cleaning, statistical analyses, and reporting-ready outputs.
- Co-drafted and supported submission of a One Health research proposal focused on artificial intelligence-enabled antimicrobial stewardship decision-making.
- Participating in an international Highly Pathogenic Avian Influenza (HPAI) modelling challenge, contributing to outbreak forecasting and intervention-scenario analyses.

Research Assistant (Part-Time) | German Federal Institute for Risk Assessment (BfR), Study Centre for Supply Chain Modelling and Artificial Intelligence (AI), Berlin (Germany)
07/2024 – 12/2025

- Packaged risk-assessment models into the FAIR Scientific Knowledge Exchange (FSKX) format; supported development of the FSKX metadata schema.
- Supported internal software development projects, including software testing; produced training materials and presentations on applied AI tools.

Research Assistant | Freie Universität Berlin, Institute for Veterinary Epidemiology and Biostatistics, Berlin (Germany)
11/2022 – 05/2025

- Coordinated coding stream and partner data specifications for the JPIAMR ENVIRE project (data templates, GitHub workflow, annual reporting); provided statistical support to junior researchers.
- Developed QMRA model to quantify human exposure and DALYs from antibiotic-resistant *E. coli* from broiler production across environmental pathways and evaluated the efficacy of interventions.
- Conducted a targeted review of manure-to-environment *E. coli* transfer; co-authored a systematic review on resistant *Enterobacteriaceae* in broilers.
- Conducted pharmacoepidemiologic (drug-utilization) analyses of more than 700,000 real-world veterinary prescribing entries on antimicrobial use (dose and indication/diagnosis), contributing to evidence for antibiotic-use monitoring in companion animals.

Doctoral Researcher | **Stiftung Tierärztliche Hochschule Hannover**, Hannover (Germany)
04/2022 - 09/2022

- Collaborated in the investigation of host-pathogen interaction of low pathogenic avian influenza, using chicken embryos as alternatives to adult animals in research models

Resident in Veterinary Public Health | **Ghent University, Veterinary Epidemiology unit**, Ghent (Belgium)
09/2020 - 03/2022

- Developed a quantitative stochastic model evaluating the economic implications of banning partial depopulation in commercial broiler production.
- Conducted environmental sampling for *Campylobacter* in poultry slaughterhouses, supporting biosecurity assessment in processing environments.
- Analyzed survey on antimicrobial stewardship and resistance education among veterinary students.

Official Veterinarian | **Netherlands Food and Consumer Product Safety Authority**, Utrecht (the Netherlands)
09/2018 - 08/2020

- Supervised animal welfare and health compliance in red meat slaughterhouses, including biosecurity and hygiene controls.
- Led a team of assistant meat inspectors and trained new veterinary officers.
- Verified identification documents and supported enforcement of EU regulatory compliance and traceability requirements.

Skills

Technical & Analytical Skills

Epidemiology, statistical analysis, evidence synthesis, risk assessment modelling, study design

Tools & Programming

R (Advanced), @RISK (Advanced), KNIME (Intermediate), Git (GitHub/GitLab), Microsoft 365 (Teams, SharePoint, Excel, PowerPoint), LLMs, reference managers

Project & Collaboration

Scientific writing, scientific communication, stakeholder engagement, interdisciplinary teamwork, project management, events organization

Languages

Italian (native), English (C1), Portuguese (B2), Dutch (B1), German (A1)

Professional Service & Committees

Treasurer, ECVPH Residents' Committee| Jan 2024-Sep 2025 | Contributed to residents' representation, communications, and event coordination

Una Europa NextVet Summer School, mentor and lecturer | June 2023 | Mentored 20 veterinary medicine students and delivered lecture on basic epidemiology and diagnostic tests

Publications (selection, full list at: https://nunziosarnino.github.io/publications.pdf)

- Quantitative microbial risk assessment of ESBL-producing *E. coli* transfer from broiler litter to lettuce, *Science in One Health*, 2026, doi.org/10.1016/j.soh.2026.100152
- Recreational Water Risk from Extended-Spectrum Beta-Lactamase-Producing *E. coli* of Broiler Origin: A Quantitative Microbial Risk Assessment, *Environments,* 2025, doi.org/10.3390/environments12110403
- Quantification of the risk of ESBL-producing *E. coli* colonization in humans through occupational exposure in broiler production, Microbial Risk Analysis, 2025, doi.org/10.1016/j.mran.2025.100349
- Use of Antibiotics in Companion Animals from 133 German Practices from 2018 to 2023, *Antibiotics*, 2025, doi.org/10.3390/antibiotics14010058
- Antibiotic Use in Horses: Analysis of 57 German Veterinary Practices (2018-2023). *Antibiotics*, 2025, doi.org/10.3390/antibiotics14090953
- Microbial risk analysis from a food industry perspective-insights from an international survey, *Microbial Risk Analysis*, 2024, doi.org/10.1016/j.mran.2024.100340
- Evaluation by Veterinary Medicine students of the education in antimicrobial stewardship and AMR at two Veterinary Medicine faculties in Belgium, *Vlaams Diergeneeskundig Tijdschrift*, 2022, doi.org/10.21825/vdt.85302
- Estimation of the Production Economic Consequences of Stopping Partial Depopulation in Broiler Production, *Animals*, 2022, doi.org/10.3390/ani12121521